

Conditional Logic, Vlookup and Xlookup

Conditional Logic

Excel logical functions - overview

Microsoft Excel provides 4 logical functions to work with the logical values. The functions are AND, OR, XOR and NOT. You use these functions when you want to carry out more than one comparison in your formula or test multiple conditions instead of just one. As well as logical operators, Excel logical functions return either TRUE or FALSE when their arguments are evaluated.

The following table provides a short summary of what each logical function does to help you choose the right formula for a specific task.

Function	Description	Formula Example	Formula Description
AND	Returns TRUE if all of the arguments evaluate to TRUE.	=AND(A2>=10, B2<5)	The formula returns TRUE if a value in cell A2 is greater than or equal to 10, and a value in B2 is less than 5, FALSE otherwise.
OR	Returns TRUE if any argument evaluates to TRUE.	=OR(A2>=10, B2<5)	The formula returns TRUE if A2 is greater than or equal to 10 or B2 is less than 5, or both conditions are met. If neither of the conditions it met, the formula returns FALSE.
XOR	Returns a logical Exclusive Or of all arguments.	=XOR(A2>=10, B2<5)	The formula returns TRUE if either A2 is greater than or equal to 10 or B2 is less than 5. If neither of the conditions is met or both conditions are met, the formula returns FALSE.

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Function	Description	Formula Example	Formula Description
NOT	Returns the reversed logical value of its argument. I.e. If the argument is FALSE, then TRUE is returned and vice versa.	=NOT(A2>=10)	The formula returns FALSE if a value in cell A1 is greater than or equal to 10; TRUE otherwise.

In additions to the four logical functions outlined above, Microsoft Excel provides 3 "conditional" functions - [IF](#), [IFERROR](#) and [IFNA](#).

Excel logical functions - facts and figures

1. In arguments of the logical functions, you can use cell references, numeric and text values, Boolean values, comparison operators, and other Excel functions. However, all arguments must evaluate to the Boolean values of TRUE or FALSE, or references or arrays containing logical values.
2. If an argument of a logical function contains any **empty cells**, such values are ignored. If all of the arguments are empty cells, the formula returns #VALUE! error.
3. If an argument of a logical function contains numbers, then zero evaluates to FALSE, and all other numbers including negative numbers evaluate to TRUE. For example, if cells A1:A5 contain numbers, the formula =AND(A1:A5) will return TRUE if none of the cells contains 0, FALSE otherwise.
4. A logical function returns the #VALUE! error if none of the arguments evaluate to logical values.
5. A logical function returns the #NAME? error if you've misspell the function's name or attempted to use the function in an earlier Excel version that does not support it. For example, the XOR function can be used in Excel 2016 and 2013 only.
6. In Excel 2007 and higher, you can include up to 255 arguments in a logical function, provided that the total length of the formula does not exceed 8,192 characters. In Excel 2003 and lower, you can supply up to 30 arguments and the total length of your formula shall not exceed 1,024 characters.

Using the AND function in Excel

The AND function is the most popular member of the logic functions family. It comes in handy when you have to test several conditions and make sure that all of them are met. Technically, the AND function tests the conditions you specify and returns TRUE if all of the conditions evaluate to TRUE, FALSE otherwise.

The syntax for the Excel AND function is as follows:

AND(logical1, [logical2], ...)

Where logical is the condition you want to test that can evaluate to either TRUE or FALSE. The first condition (logical1) is required, subsequent conditions are optional.

And now, let's look at some formula examples that demonstrate how to use the AND functions in Excel formulas.

Formula	Description
=AND(A2="Bananas", B2>C2)	Returns TRUE if A2 contains "Bananas" and B2 is greater than C2, FALSE otherwise.
=AND(B2>20, B2=C2)	Returns TRUE if B2 is greater than 20 and B2 is equal to C2, FALSE otherwise.
=AND(A2="Bananas", B2>=30, B2>C2)	Returns TRUE if A2 contains "Bananas", B2 is greater than or equal to 30 and B2 is greater than C2, FALSE otherwise.

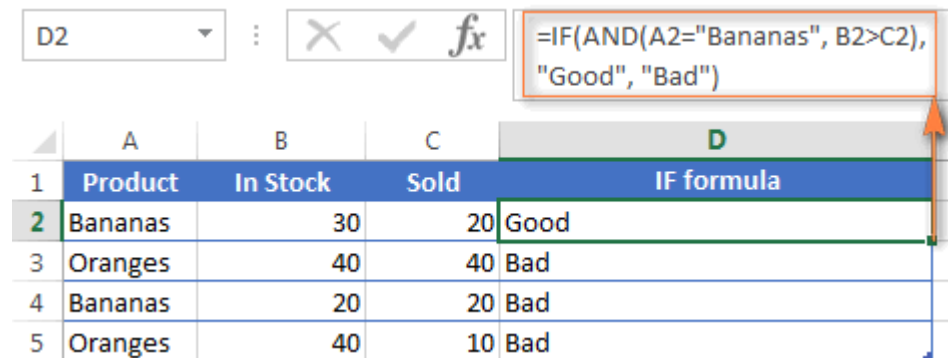
	A	B	C	D	E	F
1	Product	In Stock	Sold	Formula 1	Formula 2	Formula 3
2				=AND(A2="Bananas", B2>C1)	=AND(B2>20, B2=C2)	=AND(A2="Bananas", B2>=30, B2>C2)
3	Bananas	30	20	TRUE	FALSE	TRUE
4	Oranges	40	40	FALSE	TRUE	FALSE
5	Bananas	20	20	FALSE	FALSE	FALSE
6	Oranges	40	10	FALSE	FALSE	FALSE

Excel AND function - common uses

By itself, the Excel AND function is not very exciting and has narrow usefulness. But in combination with other Excel functions, AND can significantly extend the capabilities of your worksheets.

One of the most common uses of the Excel AND function is found in the logical_test argument of the IF function to test several conditions instead of just one. For example, you can nest any of the AND functions above inside the IF function and get a result similar to this:

=IF(AND(A2="Bananas", B2>C2), "Good", "Bad")



	A	B	C	D
1	Product	In Stock	Sold	IF formula
2	Bananas	30	20	Good
3	Oranges	40	40	Bad
4	Bananas	20	20	Bad
5	Oranges	40	10	Bad

For more IF / AND formula examples, please check out his tutorial: [Excel IF function with multiple AND conditions](#).

An Excel formula for the BETWEEN condition

If you need to create a between formula in Excel that picks all values between the given two values, a common approach is to use the IF function with AND in the logical test.

For example, you have 3 values in columns A, B and C and you want to know if a value in column A falls between B and C values. To make such a formula, all it takes is the IF function with nested AND and a couple of comparison operators:

Formula to check if X is between Y and Z, inclusive:

=IF(AND(A2>=B2,A2<=C2),"Yes", "No")

Formula to check if X is between Y and Z, not inclusive:

=IF(AND(A2>B2, A2<C2),"Yes", "No")

	Is between Values 2 and 3?	Lower- bound value	Upper- bound value	
	A	B	C	D
1	Value 1	Value 2	Value 3	Is Value 1 between Value 2 & Value 3?
2				=IF(AND(A2>=B2,A2<=C2),"Yes", "No")
3	5	1	10	Yes
4	10	6	8	No
5	4	2	5	Yes
6	12	15	3	No
7	7-Oct	5-Oct	27-Oct	Yes
8	24-Nov	26-Dec	21-Oct	No
9	13-Oct	13-Oct	17-Oct	Yes
10	Bananas	Apples	Cherries	Yes
11	Apples	Apricot	Bananas	No

As demonstrated in the screenshot above, the formula works perfectly for all data types - numbers, dates and text values. When comparing text values, the formula checks them character-by-character in the alphabetic order. For example, it states that *Apples* is not between *Apricot* and *Bananas* because the second "p" in *Apples* comes before "r" in *Apricot*. Please see [Using Excel comparison operators with text values](#) for more details.

As you see, the IF /AND formula is simple, fast and almost universal. I say "almost" because it does not cover one scenario. The above formula implies that a value in column B is smaller than in column C, i.e. column B always contains the lower bound value and C - the upper bound value. This is the reason why the formula returns "No" for row 6, where A6 has 12, B6 - 15 and C6 - 3 as well as for row 8 where A8 is 24-Nov, B8 is 26-Dec and C8 is 21-Oct.

But what if you want your between formula to work correctly regardless of where the lower-bound and upper-bound values reside? In this case, use the Excel MEDIAN function that returns the median of the given numbers (i.e. the number in the middle of a set of numbers).

So, if you replace AND in the logical test of the IF function with MEDIAN, the formula will go like:

=IF(A2=MEDIAN(A2:C2),"Yes","No")

And you will get the following results:

	A	B	C	D
1	Value 1	Value 2	Value 3	Is Value 1 between Value 2 & Value 3?
2				=IF(A2=MEDIAN(A2:C2),"Yes","No")
3	5	1	10	Yes
4	10	6	8	No
5	4	2	5	Yes
6	12	15	3	Yes
7	7-Oct	5-Oct	27-Oct	Yes
8	24-Nov	26-Dec	21-Oct	Yes
9	13-Oct	13-Oct	17-Oct	Yes
10	Bananas	Apples	Cherries	#NUM!
11	Apples	Apricot	Bananas	#NUM!

As you see, the MEDIAN function works perfectly for numbers and dates, but returns the #NUM! error for text values. Alas, no one is perfect :)

If you want a perfect Between formula that works for text values as well as for numbers and dates, then you will have to construct a more complex logical text using the AND / OR functions, like this:

=IF(OR(AND(A2>B2, A2<C2), AND(A2<B2, A2>C2)), "Yes", "No")

D2 :  =IF(OR(AND(A2>B2, A2<C2), AND(A2<B2, A2>C2)), "Yes", "No")

	A	B	C	D	E
1	Value 1	Value 2	Value 3	Is Value 1 between Value 2 and Value 3?	
2	5	1	10	Yes	
3	5-Oct	6-Oct	5-Sep	Yes	
4	Bananas	Apples	Cherries	Yes	
5	Apples	Apricot	Bananas	No	

Using the OR function in Excel

As well as AND, the Excel OR function is a basic logical function that is used to compare two values or statements. The difference is that the OR function returns TRUE if at least one of the arguments evaluates to TRUE, and returns FALSE if all arguments are FALSE. The OR function is available in all versions of Excel 2016 - 2000.

The syntax of the Excel OR function is very similar to AND:

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OR(logical1, [logical2], ...)

Where logical is something you want to test that can be either TRUE or FALSE. The first logical is required, additional conditions (up to 255 in modern Excel versions) are optional.

And now, let's write down a few formulas for you to get a feel how the OR function in Excel works.

Formula	Description
=OR(A2="Bananas", A2="Oranges")	Returns TRUE if A2 contains "Bananas" or "Oranges", FALSE otherwise.
=OR(B2>=40, C2>=20)	Returns TRUE if B2 is greater than or equal to 40 or C2 is greater than or equal to 20, FALSE otherwise.
=OR(B2=" ", C2="")	Returns TRUE if either B2 or C2 is blank or both, FALSE otherwise.

	A	B	C	D	E	F
1	Product	In Stock	Sold	Formula 1	Formula 2	Formula 3
2				=OR(A2="Bananas", A2="Oranges")	=OR(B2>=40, C2>=20)	=OR(B2="", C2="")
3	Bananas	30	10	TRUE	FALSE	FALSE
4	Oranges		20	TRUE	TRUE	TRUE
5	Cherries	20		FALSE	FALSE	TRUE
6	Oranges	30	10	TRUE	FALSE	FALSE
7	Cherries			FALSE	FALSE	TRUE

As well as Excel AND function, OR is widely used to expand the usefulness of other Excel functions that perform logical tests, e.g. the IF function. Here are just a couple of examples:

IF function with nested OR

=IF(OR(B2>30, C2>20), "Good", "Bad")

The formula returns "Good" if a number in cell B3 is greater than 30 or the number in C2 is greater than 20, "Bad" otherwise.

Excel AND / OR functions in one formula

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Naturally, nothing prevents you from using both functions, AND & OR, in a single formula if your business logic requires this. There can be infinite variations of such formulas that boil down to the following basic patterns:

=AND(OR(Cond1, Cond2), Cond3)

=AND(OR(Cond1, Cond2), OR(Cond3, Cond4))

=OR(AND(Cond1, Cond2), Cond3)

=OR(AND(Cond1,Cond2), AND(Cond3,Cond4))

For example, if you wanted to know what consignments of bananas and oranges are sold out, i.e. "In stock" number (column B) is equal to the "Sold" number (column C), the following OR/AND formula could quickly show this to you:

=OR(AND(A2="bananas", B2=C2), AND(A2="oranges", B2=C2))

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
	Product	In Stock	Sold	Supplier	Bananas & oranges sold out
2	Apples	40	30	Peter	FALSE
3	Bananas	30	20	Josh	FALSE
4	Oranges	40	40	Peter	TRUE
5	Bananas	30	20	Peter	FALSE
6	Oranges	40	10	Josh	FALSE
7	Bananas	50	50	Josh	TRUE

The formula bar shows the formula: `=OR(AND(A2="bananas", B2=C2), AND(A2="oranges", B2=C2))`



OR function in Excel conditional formatting

=OR(\$B2="", \$C2="")

The rule with the above OR formula highlights rows that contain an empty cell either in column B or C, or in both.

	A	B	C	D	E	F	G	H	I	J
1	Product	In Stock	Sold							
2	Bananas	30	20							
3	Apples	40	40							
4	Bananas		20							
5	Cherries	40	30							
6	Lemons	50								

Using the XOR function in Excel

In Excel 2013, Microsoft introduced the XOR function, which is a logical **Exclusive OR** function. This term is definitely familiar to those of you who have some knowledge of any programming language or computer science in general. For those who don't, the concept of 'Exclusive Or' may be a bit difficult to grasp at first, but hopefully the below explanation illustrated with formula examples will help.

The syntax of the XOR function is identical to OR's :

XOR(logical1, [logical2],...)

The first logical statement (Logical 1) is required, additional logical values are optional. You can test up to 254 conditions in one formula, and these can be logical values, arrays, or references that evaluate to either TRUE or FALSE.

In the simplest version, an XOR formula contains just 2 logical statements and returns:

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- TRUE if either argument evaluates to TRUE.
- FALSE if both arguments are TRUE or neither is TRUE.

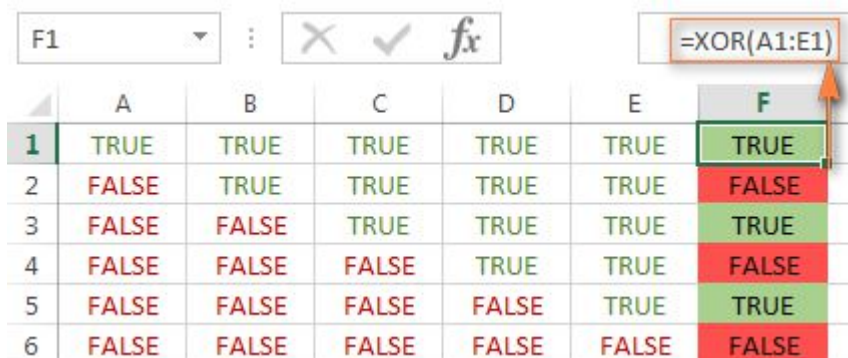
This might be easier to understand from the formula examples:

Formula	Result	Description
=XOR(1>0, 2<1)	TRUE	Returns TRUE because the 1st argument is TRUE and the 2 nd argument is FALSE.
=XOR(1<0, 2<1)	FALSE	Returns FALSE because both arguments are FALSE.
=XOR(1>0, 2>1)	FALSE	Returns FALSE because both arguments are TRUE.

When more logical statements are added, the XOR function in Excel results in:

- TRUE if an odd number of the arguments evaluate to TRUE;
- FALSE if the total number of TRUE statements is even, or if all statements are FALSE.

The screenshot below illustrates the point:



	A	B	C	D	E	F
1	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
2	FALSE	TRUE	TRUE	TRUE	TRUE	FALSE
3	FALSE	FALSE	TRUE	TRUE	TRUE	TRUE
4	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE
5	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE
6	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE

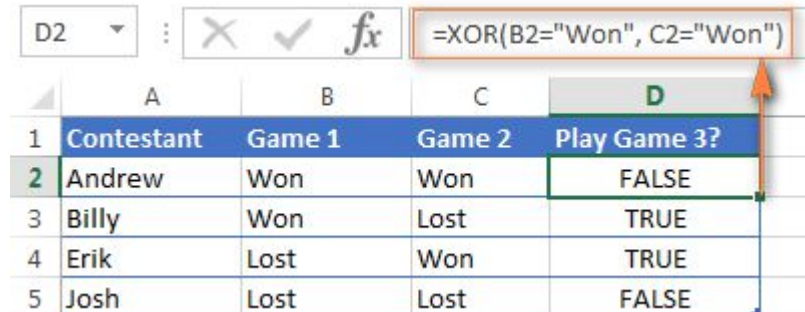
If you are not sure how the Excel XOR function can be applied to a real-life scenario, consider the following example. Suppose you have a table of contestants and their results for the first 2 games. You want to know which of the players shall play the 3rd game based on the following conditions:

- Contestants who won Game 1 and Game 2 advance to the next round automatically and don't have to play Game 3.
- Contestants who lost both first games are knocked out and don't play Game 3 either.
- Contestants who won either Game 1 or Game 2 shall play Game 3 to determine who goes into the next round and who doesn't.

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A simple XOR formula works exactly as we want:

=XOR(B2="Won", C2="Won")



	A	B	C	D
1	Contestant	Game 1	Game 2	Play Game 3?
2	Andrew	Won	Won	FALSE
3	Billy	Won	Lost	TRUE
4	Erik	Lost	Won	TRUE
5	Josh	Lost	Lost	FALSE

And if you nest this XOR function into the logical test of the IF formula, you will get even more sensible results:

=IF(XOR(B2="Won", C2="Won"), "Yes", "No")



	A	B	C	D	E
1	Contestant	Game 1	Game 2	Play Game 3?	
2	Andrew	Won	Won	No	
3	Billy	Won	Lost	Yes	
4	Erik	Lost	Won	Yes	
5	Josh	Lost	Lost	No	

Using the NOT function in Excel

The NOT function is one of the simplest Excel functions in terms of syntax:

NOT(logical)

You use the NOT function in Excel to reverse a value of its argument. In other words, if logical evaluates to FALSE, the NOT function returns TRUE and vice versa. For example, both of the below formulas return FALSE:

=NOT(TRUE)

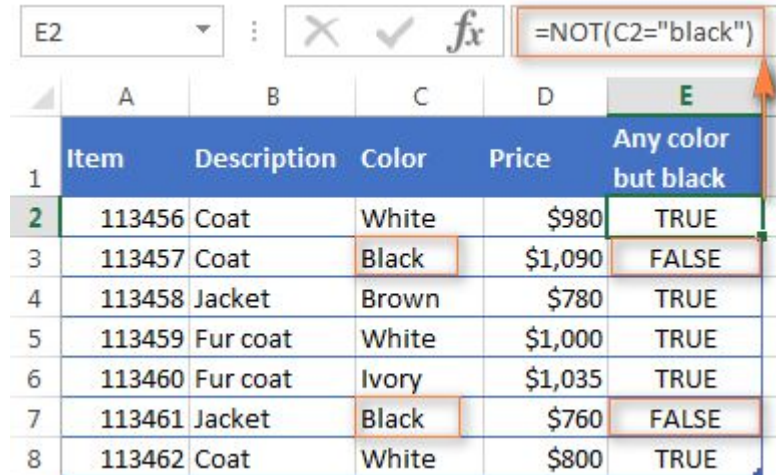
=NOT(2*2=4)

Why would one want to get such ridiculous results? In some cases, you might be more interested to know when a certain condition isn't met than when it is. For example, when reviewing a list of attire,

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you may want to exclude some color that does not suit you. I'm not particularly fond of black, so I go ahead with this formula:

=NOT(C2="black")



	A	B	C	D	E
1	Item	Description	Color	Price	Any color but black
2	113456	Coat	White	\$980	TRUE
3	113457	Coat	Black	\$1,090	FALSE
4	113458	Jacket	Brown	\$780	TRUE
5	113459	Fur coat	White	\$1,000	TRUE
6	113460	Fur coat	Ivory	\$1,035	TRUE
7	113461	Jacket	Black	\$760	FALSE
8	113462	Coat	White	\$800	TRUE

As usual, in Microsoft Excel there is more than one way to do something, and you can achieve the same result by using the [Not equal to](#) operator: **=C2<>"black"**.

If you want to test several conditions in a single formula, you can use NOT in conjunctions with the AND or OR function. For example, if you wanted to exclude black and white colors, the formula would go like:

=NOT(OR(C2="black", C2="white"))

And if you'd rather not have a black coat, while a black jacket or a black fur coat may be considered, you should use NOT in combination with the Excel AND function:

=NOT(AND(C2="black", B2="coat"))

Another common use of the NOT function in Excel is to reverse the behavior of some other function. For instance, you can combine NOT and ISBLANK functions to create the ISNOTBLANK formula that Microsoft Excel lacks.

As you know, the formula **=ISBLANK(A2)** returns TRUE if the cell A2 is blank. The NOT function can reverse this result to FALSE: **=NOT(ISBLANK(A2))**

And then, you can take a step further and create a nested IF statement with the NOT / ISBLANK functions for a real-life task:

=IF(NOT(ISBLANK(C2)), C2*0.15, "No bonus :(")

D2	:	=IF(NOT(ISBLANK(C2)), C2*0.15, "No bonus :(")			
	A	B	C	D	E
1	Salesman	Primary sales	Extra sales	Bonus	
2	Andrew	\$1,860	\$169	\$25	
3	Billy	\$910	\$145	\$22	
4	Erik	\$1,020		No bonus :(	
5	Josh	\$1,070	\$185	\$28	
6	Mike	\$1,100		No bonus :(	
7	Steve	\$1,020	\$180	\$27	

Translated into plain English, the formula tells Excel to do the following. If the cell C2 is not empty, multiply the number in C2 by 0.15, which gives the 15% bonus to each salesman who has made any extra sales. If C2 is blank, the text "No bonus :(" appears.

In essence, this is how you use the logical functions in Excel. Of course, these examples have only scratched the surface of AND, OR, XOR and NOT capabilities. Knowing the basics, you can now extend your knowledge by tackling your real tasks and writing smart elaborate formulas for your worksheets.

Countif() and countifs()

The COUNTIF function in Excel counts the number of cells that meet a specific criteria. The syntax for COUNTIF is:

- **COUNTIF(range, criteria)**
- **range:** The group of cells to count
- **criteria:** A number, expression, cell reference, or text string that determines which cells to count

The COUNTIFS function counts the number of cells in a range that meet one or more conditions. It is an advanced version of the COUNTIF function, which only allows for one criteria.

- **COUNTIFS(range1, criteria1, range2, criteria2, ...)**
- **range1:** The first range of cells to evaluate.
- **criteria1:** The criteria or condition to be met in range1.
- **range2:** (Optional) The second range of cells to evaluate.
- **criteria2:** The criteria or condition to be met in range1.

Sumif() and sumifs()

The SUMIF function in Excel calculates the sum of values in a range based on a true or false condition.

The syntax for the SUMIF function in Excel is =SUMIF(range, criteria, [sum_range]):

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- **range:** The cells to evaluate against the criteria
- **criteria:** The condition that the cells in the range must meet
- **sum_range:** The range of cells to sum if the corresponding cells in the range meet the criteria

The SUMIFS function calculates the sum of values in a range that meet multiple criteria.

To use the SUMIFS function in Excel with multiple criteria, you can use the following syntax:

SUMIFS(sum_range, criteria_range1, criteria1, [criteria_range2, criteria2], ...)

The first three arguments are required:

- **sum_range**: The cells to sum
- **criteria_range1**: The range of cells where the criteria are located
- **criteria1**: The criteria to be found in criteria_range1

You can set up to 255 conditions with SUMIFS

Note that the sum_range is required as the first argument as the number of criteria is variable.

Other functions that have the similar syntax include:

- **AveragelFs**: Averages based on one or more criteria
- **MaxlFs**: Finds the largest value in a set based on one or more criteria
- **MinlFs**: Finds the smallest value in a set based on one or more criteria



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Vlookup() and Xlookup()

The VLOOKUP function is used to find a value in a table or a range by row.

In its simplest form, the VLOOKUP function says:

The syntax for VLOOKUP is as follows:

=VLOOKUP(look_up_value, array, col_no,, TRUE)

- **look_up_value** may be a cell reference or a fixed value.
- **array** is the range of cells in which to search for the value you want to find.
- **Col_no.** is the column in that range of cells that contains the value that you seek.

Tips:

- The secret to VLOOKUP is to organize your data so that the value you look up is to the left of the return value you want to find.

How to get started

There are four pieces of information that you will need in order to build the VLOOKUP syntax:

1. The value you want to look up, also called the lookup value.
2. The range where the lookup value is located. Remember that the lookup value should always be in the first column in the range for VLOOKUP to work correctly. For example, if your lookup value is in cell C2 then your range should start with C.
3. The column number in the range that contains the return value. For example, if you specify B2:D11 as the range, you should count B as the first column, C as the second, and so on.
4. Optionally, you can specify TRUE if you want an approximate match or FALSE if you want an exact match of the return value. If you don't specify anything, the default value will always be TRUE or approximate match.

Examples

Here are a few examples of VLOOKUP:



Example 1

	A	B	C	D	E
1	ID	Last name	First name	Title	Birth date
2	101	Davis	Sara	Sales Rep	12/08/68
3	102	Fontana	Olivier	VP (Sales)	02/19/52
4	103	Leal	Karina	Sales Rep	08/30/63
5	104	Patten	Michael	Sales Rep	09/19/58
6	105	Burke	Brian	Sales Manager	03/04/55
7	106	Sousa	Luis	Sales Rep	07/02/63
8					
9					
10	Formula	=VLOOKUP(B3,B2:E7,2,FALSE)			
11	Result	Olivier			
12					

VLOOKUP looks for *Fontana* in the first column (column B) in table_array B2:E7, and returns *Olivier* from the second column (column C) of the table_array. FALSE returns an exact match.

Example 2

	A	B	C	D	E
1	ID	Last name	First name	Title	Birth date
2	101	Davis	Sara	Sales Rep	12/08/68
3	102	Fontana	Olivier	VP (Sales)	02/19/52
4	103	Leal	Karina	Sales Rep	08/30/63
5	104	Patten	Michael	Sales Rep	09/19/58
6	105	Burke	Brian	Sales Manager	03/04/55
7	106	Sousa	Luis	Sales Rep	07/02/63
8					
9					
10	Formula	=VLOOKUP(102,A2:C7,2,FALSE)			
11	Result	Fontana			

VLOOKUP looks for an exact match (FALSE) of the last name for 102 (lookup_value) in the second column (column B) in the A2:C7 range, and returns *Fontana*.



Example 3

	A	B	C	D	E
1	ID	Last name	First name	Title	Birth date
2	101	Davis	Sara	Sales Rep	12/08/68
3	102	Fontana	Olivier	VP (Sales)	02/19/52
4	103	Leal	Karina	Sales Rep	08/30/63
5	104	Patten	Michael	Sales Rep	09/19/58
6	105	Burke	Brian	Sales Manager	03/04/55
7	106	Sousa	Luis	Sales Rep	07/02/63
8					
9					
10	Formula	=IF(VLOOKUP(103,A1:E7,2,FALSE)="Sousa","Located","Not found")			
11	Result	Not found			

IF checks to see if VLOOKUP returns *Sousa* as the last name of employee corresponding to 103 (lookup_value) in A1:E7 (table_array). Because the last name corresponding to 103 is *Leal*, the IF condition is false, and *Not found* is displayed.

Example 4

	A	B	C	D	E
1	ID	Last name	First name	Title	Birth date
2	101	Davis	Sara	Sales Rep	12/08/68
3	102	Fontana	Olivier	VP (Sales)	02/19/52
4	103	Leal	Karina	Sales Rep	08/30/63
5	104	Patten	Michael	Sales Rep	09/19/58
6	105	Burke	Brian	Sales Manager	03/04/55
7	106	Sousa	Luis	Sales Rep	07/02/63
8					
9					
10	Formula	=INT(YEARFRAC(DATE(2014,6,30), VLOOKUP(105,A2:E7,5, FALSE), 1))			
11	Result	59			
12					
13					
14					
15					

VLOOKUP looks for the birth date of the employee corresponding to 105 (lookup_value) in the A2:E7 range (table_array), and returns 03/04/1955. Then, YEARFRAC subtracts this birth date from 2014/6/30 and returns a value, which is then converted by INT to the integer 59.

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Example 5

	A	B	C	D	E
1	ID	Last name	First name	Title	Birth date
2	101	Davis	Sara	Sales Rep	12/08/68
3	102	Fontana	Olivier	VP (Sales)	02/19/52
4	103	Leal	Karina	Sales Rep	08/30/63
5	104	Patten	Michael	Sales Rep	09/19/58
6	105	Burke	Brian	Sales Manager	03/04/55
7	106	Sousa	Luis	Sales Rep	07/02/63
8					
9					
10	Formula	=IF(ISNA(VLOOKUP(105,A2:E7,2,FALSE)) = TRUE, "Employee not found", VLOOKUP(105,A2:E7,2,FALSE))			
11	Result	Burke	IF checks to see if VLOOKUP returns a value for last name from column B for 105 (lookup_value). If VLOOKUP finds a last name, then IF will display the last name, otherwise IF returns <i>Employee not found</i>. ISNA makes sure that if VLOOKUP returns #N/A, then the error is replaced by <i>Employee not found</i>, instead of #N/A. In this example, the return value is <i>Burke</i>, which is the last name corresponding to 105.		
12					
13					
14					

The XLOOKUP() function is a more flexible function that does not require the organising of the data in a certain order for the result.

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

Argument	Description
lookup_value	The value to search for
Required*	*If omitted, XLOOKUP returns blank cells it finds in lookup_array .
lookup_array	The array or range to search
Required	

Argument	Description
return_array	The array or range to return
Required	
[if_not_found]	Where a valid match is not found, return the [if_not_found] text you supply.
Optional	If a valid match is not found, and [if_not_found] is missing, #N/A is returned.
[match_mode]	Specify the match type:
Optional	0 - Exact match. If none found, return #N/A. This is the default. -1 - Exact match. If none found, return the next smaller item. 1 - Exact match. If none found, return the next larger item. 2 - A wildcard match where *, ?, and ~ have special meaning.
[search_mode]	Specify the search mode to use:
Optional	1 - Perform a search starting at the first item. This is the default. -1 - Perform a reverse search starting at the last item. 2 - Perform a binary search that relies on lookup_array being sorted in <i>ascending</i> order. If not sorted, invalid results will be returned. -2 - Perform a binary search that relies on lookup_array being sorted in <i>descending</i> order. If not sorted, invalid results will be returned.

Examples

Example 1 uses XLOOKUP to look up a country name in a range, and then return its telephone country code. It includes the **lookup_value** (cell F2), **lookup_array** (range B2:B11), and **return_array** (range D2:D11) arguments. It doesn't include the **match_mode** argument, as XLOOKUP produces an exact match by default.

G2						
=XLOOKUP(F2,B2:B11,D2:D11)						
	A	B	C	D	E	F
1		Country	Abr	Prefix		What is the dial code?
2		China	CN	+86		Brazil
3		India	IN	+91		
4		United States	US	+1		
5		Indonesia	ID	+62		
6		Brazil	BR	+55		
7		Pakistan	PK	+92		
8		Nigeria	NG	+234		
9		Bangladesh	BD	+880		
10		Russia	RU	+7		
11		Mexico	MX	+52		

Note: XLOOKUP uses a lookup array and a return array, whereas VLOOKUP uses a single table array followed by a column index number. The equivalent VLOOKUP formula in this case would be: **=VLOOKUP(F2,B2:D11,3,FALSE)**

Example 2 looks up employee information based on an employee ID number. Unlike VLOOKUP, XLOOKUP can return an array with multiple items, so a single formula can return both employee name and department from cells C5:D14.

C2			
=XLOOKUP(B2,B5:B14,C5:D14)			
	A	B	C
1		Emp ID	Employee Name
2		8389	Dianne Pugh
3			
4		Emp ID	Employee Name
5		4390	Ned Lanning
6		8604	Margo Hendrix
7		8389	Dianne Pugh
8		4937	Earlene McCarty
9		8299	Mia Arnold
10		2643	Jorge Fellows
11		5243	Rose Winters
12		9693	Carmela Hahn
13		1636	Delia Cochran
14		6703	Marguerite Cervantes

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Example 3 adds an **if_not_found** argument to the preceding example.

C2				=XLOOKUP(B2,B5:B14,C5:D14,"ID not found")			
	A	B	C	D			
1		Emp ID	Employee Name	Department			
2		1234	ID not found				
3							
4		Emp ID	Employee Name	Department			
5		4390	Ned Lanning	Marketing			
6		8604	Margo Hendrix	Sales			
7		8389	Dianne Pugh	Finance			
8		4937	Earlene McCarty	Accounting			
9		8299	Mia Arnold	Operations			
10		2643	Jorge Fellows	Executive			
11		5243	Rose Winters	Sales			
12		9693	Carmela Hahn	Finance			
13		1636	Delia Cochran	Accounting			
14		6703	Marguerite Cervantes	Marketing			

Example 4 looks in column C for the personal income entered in cell E2, and finds a matching tax rate in column B. It sets the **if_not_found** argument to return 0 (zero) if nothing is found. The **match_mode** argument is set to 1, which means the function will look for an exact match, and if it can't find one, it returns the next larger item. Finally, the **search_mode** argument is set to 1, which means the function will search from the first item to the last.

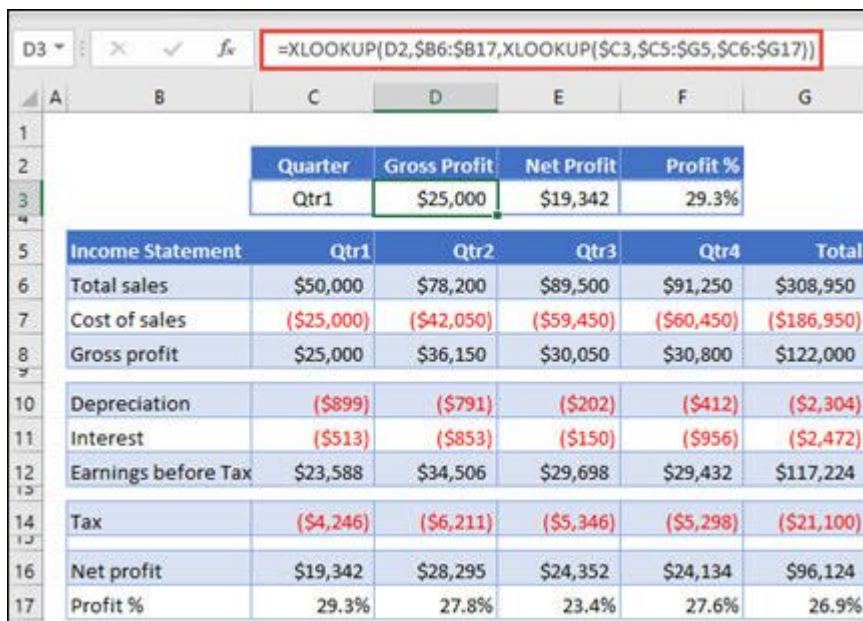
F2						=XLOOKUP(E2,C2:C7,B2:B7,0,1,1)					
	A	B	C	D	E	F					
1		Tax Rate	Max Income		Income	Tax Rate					
2		10%	\$9,700		\$46,523	24%					
3		22%	\$39,475								
4		24%	\$84,200								
5		32%	\$160,726								
6		35%	\$204,100								
7		37%	\$510,300								

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Note: XARRAY's **lookup_array** column is to the right of the **return_array** column, whereas VLOOKUP can only look from left-to-right.

Example 5 uses a nested XLOOKUP function to perform both a vertical and horizontal match. It first looks for **Gross Profit** in column B, then looks for **Qtr1** in the top row of the table (range C5:F5), and finally returns the value at the intersection of the two. This is similar to using the [INDEX](#) and [MATCH](#) functions together.

Tip: You can also use XLOOKUP to replace the [HLOOKUP](#) function.



	Quarter	Gross Profit	Net Profit	Profit %
Qtr1		\$25,000	\$19,342	29.3%

Income Statement	Qtr1	Qtr2	Qtr3	Qtr4	Total
Total sales	\$50,000	\$78,200	\$89,500	\$91,250	\$308,950
Cost of sales	(\$25,000)	(\$42,050)	(\$59,450)	(\$60,450)	(\$186,950)
Gross profit	\$25,000	\$36,150	\$30,050	\$30,800	\$122,000
Depreciation	(\$899)	(\$791)	(\$202)	(\$412)	(\$2,304)
Interest	(\$513)	(\$853)	(\$150)	(\$956)	(\$2,472)
Earnings before Tax	\$23,588	\$34,506	\$29,698	\$29,432	\$117,224
Tax	(\$4,246)	(\$6,211)	(\$5,346)	(\$5,298)	(\$21,100)
Net profit	\$19,342	\$28,295	\$24,352	\$24,134	\$96,124
Profit %	29.3%	27.8%	23.4%	27.6%	26.9%

Note: The formula in cells D3:F3 is: **=XLOOKUP(D2,\$B6:\$B17,XLOOKUP(\$C3,\$C5:\$G5,\$C6:\$G17))**.

Reference Links

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